

Memory Control Signals Emerge Before Action in Long Horizon Agents

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ABSTRACT

Long horizon language model agents continuously accumulate interaction history, increasing computational cost while making relevant information harder to preserve and reuse. Existing context management methods mainly focus on how to compress or retrieve history, but largely leave open whether the model itself already represents the need for these memory operations before they occur. We study the hidden state immediately before each agent action and find that compression and recall needs are already encoded in the model’s internal representations. These signals cannot be explained by simple context length or interaction progress, and they exhibit distinct formation patterns across model depth. We further show that most memory decision information is preserved in a compact recent context, while selectively restored historical evidence complements the long range dependencies that recent context misses. Based on these findings, we propose **Preaction Memory with Evidence Retrieval(PaMER)**, which combines state guided compression with external evidence retrieval. PaMER+ further introduces step level evidence selection to recover only the historical information required by the current task. Experiments on WorkBuddyBench, across multiple context management baselines and model backbones, show that our framework substantially reduces context consumption while maintaining competitive task performance.

1 Introduction

Long horizon language model agents face a direct challenge: interaction history continuously grows, while longer context increases computation, latency, and memory cost and may also reduce the model’s ability to use relevant information effectively^{1,2,3}. A common strategy is to compress history when the context approaches its window limit, or to trigger compression periodically according to token count or interaction length. However, context length is not equivalent to compression need. Two agent trajectories with the same number of tokens may represent very different task states. One may be long because the agent has just assembled critical evidence, while another may already contain repeated searches, irrelevant tool outputs, and outdated intermediate results. Recent work therefore begins to let agents manage context according to the current task state rather than relying only on fixed length rules^{4,5,6,7,8}.

Compression introduces a complementary problem: information removed from the active context may become important again later. When subsequent reasoning depends on earlier evidence,

the agent must recover or retrieve historical information^{9,10,11,5,12,13,14}. Long horizon memory management therefore involves two dynamic decisions: whether the current history should be compressed, and whether earlier information is needed again. Existing methods implement such operations through rules, prompting, additional controllers, or dedicated training^{15,4,5,9,16,17,18,19}. This leaves a more fundamental issue unresolved: whether the language model already represents its memory need internally before compression or historical recall is performed.

We study the hidden state immediately before the model produces its next action at each agent decision point. Our hypothesis is that this preaction hidden state has already integrated the task objective, recent evidence, and historical dependencies, and can therefore reveal whether the current history should be compressed or whether earlier evidence is needed again. The final preaction hidden state predicts compression and recall with AUROC values of **0.831** and **0.765**, respectively, substantially outperforming observable controls based on context length, turn count, and tool type. More importantly, we further investigate **how this information forms inside the model**. Layerwise analysis shows that compression information generally becomes stronger with model depth, whereas recall information emerges earlier and reaches its highest measured performance at an intermediate layer. The representation at the last input token also consistently outperforms mean pooling. These results suggest that the model contains more than a simple signal that the context has become long. Instead, memory needs are reflected in internal representations that depend on the current task state and develop differently during model computation.

We next study how much history is required to preserve this decision information. Retaining only the task prefix and the two most recent complete interaction blocks uses approximately 26% of the Full Context input, yet preserves most of the predictive performance for both compression and recall. This indicates that the recent working state already contains most of the dynamic information required for memory decisions. Recent context alone, however, is not sufficient. When long range dependencies become relevant again, restoring a small amount of selected raw historical evidence recovers additional predictive information and outperforms the hidden state memory alternatives that we evaluate. This leads to a natural design principle: recent context describes the current working state, while external history restores the evidence that is currently missing.

Based on these findings, we propose **PaMER**, Preaction Memory with Evidence Retrieval. PaMER uses the preaction representation extracted by a shared frozen Qwen3.5-9B model to decide when to compress, removes older history from the active context, and stores it in retrievable external memory for later task-specific evidence reuse. PaMER+ further introduces step level evidence selection so that only the raw historical steps required by the current task are restored.

Our contributions are threefold. **First**, we systematically study whether compression and recall needs are already present in the model’s internal representations before the corresponding memory operations occur. We show that preaction hidden states contain memory decision information that cannot be explained by simple context length or interaction progress, and we further characterize how these signals form across model depth. **Second**, we investigate which parts of history support these decisions. Approximately 26% of Full Context is sufficient to preserve most predictive performance, while selective historical evidence recovers long range dependencies that are missing from recent context. **Third**, we translate these findings into PaMER and PaMER+, and evaluate them on WorkBuddyBench²⁰ against multiple context management baselines and across several model backbones. The resulting system substantially reduces context consumption while maintaining competitive task performance.

2 Related Work

2.1 Context Compression for Long Horizon Agents

Long context processing has motivated a broad class of methods that reduce the amount of history presented to a language model while preserving task relevant information. LLMLingua 2 learns token importance through distilled supervision, while LongLLMLingua extends prompt compression to long context settings^{21,22}. As language models are increasingly used as agents, recent work has moved from static prompt compression to the management of continuously growing interaction histories. ACON¹⁵ optimizes compression for long horizon agent trajectories, SWE Pruner²³ adaptively removes redundant context in coding agents, and PACE²⁴ selects historical information according to its predicted relevance to future actions.

Recent studies further suggest that compression timing should depend on the current task state rather than only on a fixed context threshold. Self Compact allows an agent to decide when to compact its trajectory, Self GC continuously reorganizes context during interaction, and Sculptor provides explicit operations for summarizing, hiding, restoring, and searching information^{4,25,5}. State Proprioception similarly exposes runtime information to support adaptive context management²⁶. These methods show that context length alone does not fully characterize when memory intervention is useful. Our work studies an earlier question in this process. Rather than first providing an explicit context state or training a new memory policy, we ask whether the model’s own hidden representation already contains information about the need to compress.

2.2 Memory Retrieval and Historical Evidence

Removing information from active context creates a complementary problem: historical evidence may become useful again later^{27,28,29,30,31}. MemGPT organizes information between active and external memory, SAM adapts memory access to the current reasoning state, CoMem introduces a separate long context component for context management, and ACM studies agentic context management for long horizon tasks^{32,9,10,11}. Learning What Not to Forget further studies lightweight mechanisms for deciding which parts of long agent histories should remain available^{33,34,35}.

Retrieval itself has also been studied as a dynamic decision. FLARE performs retrieval according to signals observed during generation, while Self RAG learns when to retrieve and how to use retrieved evidence through explicit reflection^{36,37,38,39}. Most closely related to our analysis, Probing RAG uses internal representations to determine whether additional document retrieval is necessary⁴⁰. Our setting differs in both the target and the interaction structure. Probing RAG studies selective retrieval of external documents for question answering, whereas we study two memory needs inside long agent trajectories: whether accumulated interaction should be compressed and whether evidence that previously left the active context is needed again. We further examine how these two signals develop across model depth and how much of the trajectory history is required to preserve them.

2.3 Probing Internal Representations

Our analysis is also related to work that uses probes to study information encoded in neural representations. Linear probes provide a simple way to measure whether a target property can be decoded from intermediate model states⁴¹. At the same time, prior work emphasizes that probe accuracy should be interpreted carefully and does not by itself identify the causal mechanism used by the underlying model^{42,43}.

Related studies have shown that hidden representations can expose information that is not directly available from model outputs. Such signals include latent knowledge⁴⁴, confidence about whether an answer is known^{45,46}, and information related to truthfulness⁴⁷. We use probing for a more specific systems question: whether the model state immediately before an agent action already contains information about an upcoming memory need. Accordingly, our claims are predictive rather than causal. We compare hidden states with observable metadata and surface text controls, analyze how compression and recall signals evolve across layers, and study how these signals change when most historical context is removed. These findings then provide the empirical basis for the design of our online memory controller for long horizon agent interactions.

3 Understanding Memory Needs in Long Horizon Agents

We study whether compression and recall needs are represented internally before the model produces its next action. Our analysis defines these memory decisions from long agent trajectories, extracts the corresponding preaction hidden states, and measures how well the decisions can be read from different model layers across successive stages of computation.

We first compare hidden states with observable controls, then examine how the signal changes across layers. Finally, we vary the available history to test how much decision information remains under a smaller context budget and after historical evidence is restored.

3.1 Memory Decision Data and Probing

We collect 72,912 decision points from 2,521 long agent trajectories covering 600 questions. The source questions do not overlap with WorkBuddyBench and come from different task domains. An LLM annotator identifies states where compression or recall is needed. The annotation procedure and the use of multiple labeled points within one trajectory are described in Appendix A.1.

At decision step t , let o_t denote the context available before the next action. For trajectory τ , the annotator identifies compression and recall decision sets $\mathcal{A}^c(\tau)$ and $\mathcal{A}^r(\tau)$. We define

$$y_t^m = \mathbf{1}[t \in \mathcal{A}^m(\tau)], \quad m \in \{c, r\}, \quad (1)$$

where c denotes compression and r denotes recall. Compression indicates that older context should be condensed while preserving task relevant information. Recall indicates that the current reasoning state needs earlier evidence that is no longer available in the active context. Recall is evaluated only when valid external memory exists. A trajectory may contain multiple positive decision points because the same information need can arise again during a long interaction. This formulation allows memory needs to be evaluated at the decision level while preserving their dependence on the surrounding trajectory context.

For each decision point, we extract the model state immediately before the next action. Let $\phi^{(L)}(o_t)$ denote the final hidden sequence after the output RMSNorm. Our main representation is

$$\mathbf{h}_t = \phi^{(L)}(o_t)[-1] \in \mathbb{R}^d, \quad d = 4096. \quad (2)$$

The input uses the same conversation format and tool descriptions as the corresponding agent state. Neither the annotation nor the future action is included when extracting $\mathbf{h}_t$.

We train an independent readout for each memory decision. The linear probe predicts

$$p_t^m = \sigma((\mathbf{w}^m)^\top \mathbf{h}_t + b^m), \quad m \in \{c, r\}. \quad (3)$$

We additionally evaluate a one hidden layer MLP to test whether nonlinear readout provides additional predictive power. The language model itself is not updated during probe training. This separation is intended to test what information is already accessible from the pretrained representation without adapting the underlying model.

Because both targets are imbalanced, we optimize class weighted binary cross entropy,

$$\mathcal{L}_m = -\frac{1}{N_m} \sum_{i=1}^{N_m} [\rho_m y_i^m \log p_i^m + (1 - y_i^m) \log(1 - p_i^m)], \quad (4)$$

where $\rho_m = N_m^-/N_m^+$ is computed from the training split only. All states from the same question remain in the same split. We use 70/15/15 training, validation, and test splits over five seeds. Decision thresholds are selected by validation F1 and then fixed for testing. AUROC is the primary metric. AUPRC, thresholded metrics, and the complete MLP definition are reported in Appendix A.2.

This evaluation protocol treats each question as the grouping unit rather than individual decision points, reducing leakage between closely related states from the same underlying task. It also ensures that probe performance reflects generalization across questions rather than memorization of trajectory-specific patterns. This setting more closely reflects deployment, where the controller must generalize to unseen tasks. It also provides a stricter test of whether the learned signal transfers beyond the trajectories used for probe fitting.

To characterize how memory decision information develops across model depth, we train an independent linear probe at each evaluated layer. At layer ℓ , we compare the representation at the final input position with the mean representation over all input tokens:

$$\mathbf{h}_{t,\text{last}}^{(\ell)} = \phi^{(\ell)}(o_t)[-1], \quad \mathbf{h}_{t,\text{mean}}^{(\ell)} = \frac{1}{n_t} \sum_{j=1}^{n_t} \phi^{(\ell)}(o_t)[j]. \quad (5)$$

For each layer, target, and pooling choice, we also measure the difference between the mean positive and negative probe logits,

$$\Delta_{\ell,\alpha}^m = \mathbb{E}[z_{t,\ell,\alpha}^m \mid y_t^m = 1] - \mathbb{E}[z_{t,\ell,\alpha}^m \mid y_t^m = 0], \quad (6)$$

where $\alpha \in \{\text{last}, \text{mean}\}$. AUROC remains the primary metric because the magnitude of the logit gap also depends on probe scale.

3.2 Memory Signals beyond Observable Context

A simple explanation for memory decisions is that the agent compresses because the context is long, or recalls because the interaction has lasted many turns. Table 1 shows that these observable signals are informative but insufficient. Metadata reaches 0.758 AUROC for compression and 0.549 for recall, whereas the final hidden state reaches 0.831 and 0.765, respectively. A nonlinear MLP provides almost no additional gain, suggesting that much of this information is already accessible through a linear readout.

Table 1: Memory decision signals. Mean AUROC.

Signal	Comp. $\uparrow$	Recall $\uparrow$
Length / Turn	0.751	0.552
Metadata	0.758	0.549
Hidden (Linear)	0.831	0.765
Hidden (MLP)	0.827	0.763

These results show that preaction hidden states contain predictive information beyond the tested metadata. They do not imply that the probes identify the model’s causal mechanism. Additional text controls, uncertainty estimates, and robustness results are reported in Appendix A.

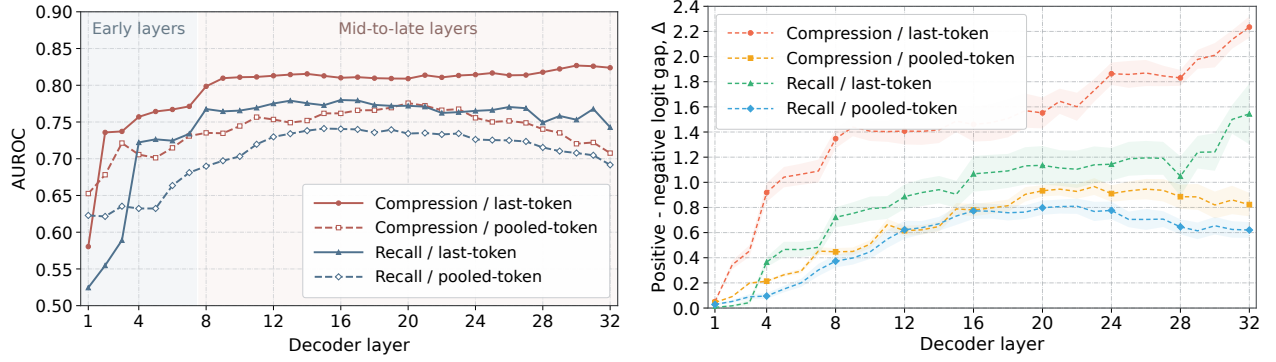


Figure 1: **Memory-decision signals across layers.** (a) AUROC of independently trained probes using the final input token or mean-pooled states. (b) Mean positive-minus-negative probe logit. Results are aggregated over five question-grouped splits.

3.3 How Memory Signals Form across Depth

Figure 1 shows that compression prediction becomes stronger with model depth. The earliest measured representation provides only a weak signal, with AUROC around 0.57, whereas the final normalized state reaches 0.831. Recall follows a different pattern and reaches its highest measured mean AUROC of 0.780 at block 16, compared with 0.765 at the final state. The two memory needs therefore develop differently across the network.

The final input position is consistently more informative than mean pooling. At the final state, last-token and mean-pooled AUROCs are 0.831 and 0.712 for compression, and 0.765 and 0.696 for recall. This suggests that information relevant to the next memory action is more accessible near the position used to produce that action than in an average over the full input.

Although recall peaks at block 16, we use the final state for the online controller. Compression, which is the learned online decision in PAMER, improves from 0.810 at block 16 to 0.831 at the final state. The recall advantage at block 16 is 0.015, while the reported variation across splits is larger than this difference. More importantly, the final preaction state provides a consistent interface across models with different depths, whereas a fixed intermediate block does not have the same meaning across architectures. We therefore use the final state as the controller readout for stable cross-model deployment behavior without claiming that it is the best layer for every memory-related target. Exact layer-level results are reported in Appendix Table 9.

3.4 What Historical Information Is Needed

We next reduce the amount of trajectory history available to the model while keeping the prediction targets unchanged. Let P_t denote the system and task prefix and let $\mathcal{R}_t$ contain the two most recent complete interaction blocks. The Recent2 condition uses

$$C_t^{\text{recent}} = P_t \oplus \mathcal{R}_t. \quad (7)$$

This construction preserves the task objective together with the current working state while removing most earlier interaction. Recent2 uses only 26.0% of the Full Context token budget, yet retains compression and recall AUROCs of 0.776 and 0.761, compared with 0.829 and 0.780 under Full Context. Decision distillation keeps the same context budget. It changes compression AUROC only from 0.776 to 0.781, but increases compression F1 from 0.541 to 0.598, suggesting that Recent2

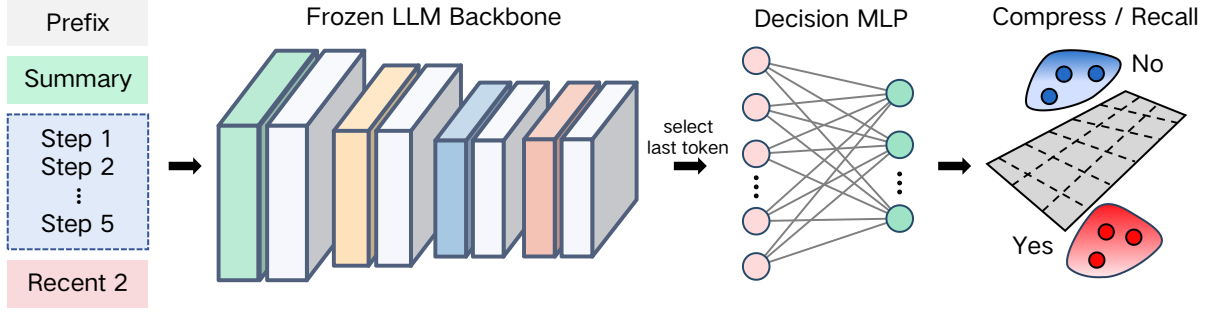


Figure 2: **Reading memory needs before an action.** The diagram illustrates compression and recall as prediction targets. In the deployed controller, the compression heads read the final state from the task prefix and Recent2. The recall probe is used for analysis, not as an online retrieval gate. already contains much of the required information while the Full Context teacher mainly improves the thresholded decision.

The remaining gap shows that recency alone is not sufficient. Some dependencies needed for a memory decision lie outside the recent window. If $\mathcal{E}_t$ denotes recovered historical evidence, the reconstructed context is

$$C_t^{\text{recover}} = P_t \oplus \mathcal{E}_t \oplus \mathcal{R}_t. \quad (8)$$

The task and system prefix is retained in every condition, so all comparisons preserve the goal against which relevance is judged. The prefix-only Current condition is reported in Appendix B. These comparisons do not isolate the causal contribution of the task prefix.

Table 2 shows that a short recent history preserves most of the recall signal and a large fraction of the compression signal. However, the form of older information also matters. Hidden Bank Recent5 uses the smallest token budget in the table but loses more decision information. Restoring selected raw evidence is more effective in this setting: Selective Raw Top5 reaches 0.832 compression AUROC and 0.769 recall AUROC. Qwen Top3 provides a practical embedding-based alternative, reaching 0.804 and 0.764 with 53.9% of the Full Context token budget, with a moderate token budget.

Taken together, the results suggest three complementary sources of information. The task prefix establishes what is relevant, recent interaction describes the current working state, and selected historical evidence restores dependencies that have left the recent window. We do not interpret this as a causal decomposition. Instead, it provides the design pattern used by PAMER in the next section. Full context sweeps and retrieval controls are reported in Appendix B.

Table 2: Compact context and historical evidence. Tokens are relative to Full Context.

History	Tokens ↓	Comp. ↑	Recall ↑
Full Context	100.0	0.829	0.780
<i>Compact context</i>			
Recent2	26.0	0.776	0.761
+ Decision KD	26.0	0.781	0.753
<i>Recovered history</i>			
Hidden Bank	16.8	0.769	0.730
Selective Raw	66.2	0.832	0.769
Qwen Retrieval	53.9	0.804	0.764

4 PAMER: Preaction Memory with Evidence Retrieval

PAMER uses the current model state to decide when to compress history and an external memory to retain information for later access. PAMER⁺ adds selection at the level of individual historical steps. The design follows the preceding analysis: a short recent context can support compression decisions, while selected older evidence remains available when information has left the active history.

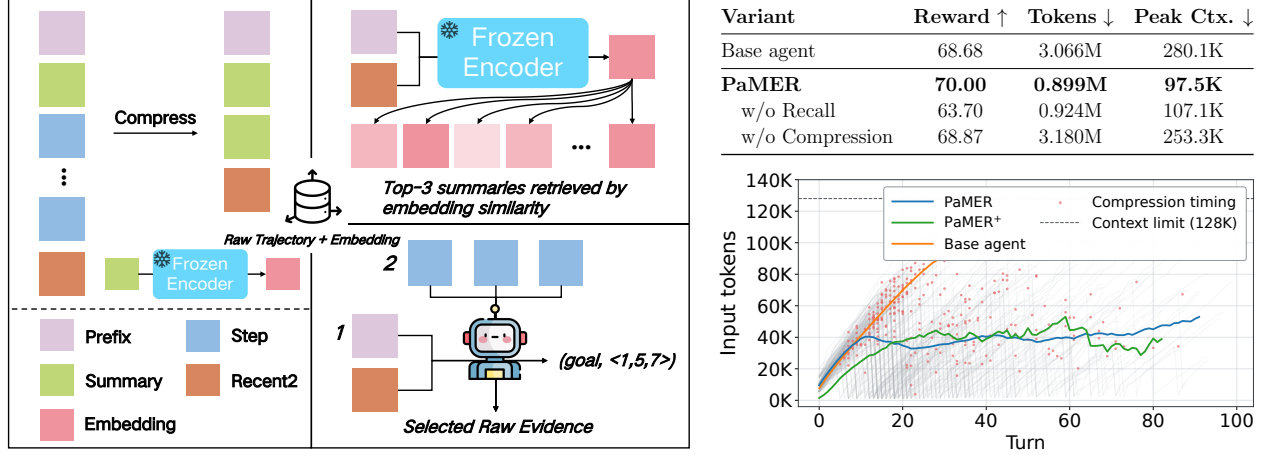


Figure 3: **Online behavior and component comparisons.** (a) Compression and evidence recall. (b) Component results. (c) Input context across interaction turns. The horizontal line marks 128K as an analysis reference, not a hard limit enforced by the controller.

4.1 State Guided Compression and Retrieval

PAMER retains the system and task prefix P_t and the two most recent complete interaction blocks, $\mathcal{R}_t$. A block contains an assistant message and its subsequent tool results and other messages, up to the next assistant message. Earlier complete blocks form the compression candidate. Operating on complete blocks preserves the assistant and tool-call structure.

For a candidate of at least 1,024 tokens, five independently trained compression heads read the final state from $P_t \oplus \mathcal{R}_t$. Compression is triggered when at least three heads exceed their validation-selected thresholds. The heads are trained on Recent2 using both the compression annotation and the corresponding Full Context teacher prediction:

$$\mathcal{L}_s = \text{WBCE}(z_{t,s}, y_t^c) + 0.5 \text{ BCEWithLogits}(z_{t,s}, p_{t,s}^{\text{full}}). \quad (9)$$

Here s indexes the independently trained head. The Full Context teacher is used during training, not during online decisions.

When compression is triggered, we summarize the older blocks. Each summary is encoded with Qwen3-Embedding-8B and stored with links to the covered history. The raw blocks are removed only after storage succeeds. The current request then contains

$$C'_t = P_t \oplus \mathcal{R}_t. \quad (10)$$

The new summary is not inserted into this same request, and no recall is performed on it. This prevents immediate retrieval from undoing the context reduction.

On a later request with available memory, the task and Recent2 form an embedding query. The controller retrieves the Top3 summaries by cosine similarity and inserts them between the prefix and recent blocks. The current implementation does not use the learned recall probe as a gate. Thus, recall probing and online retrieval remain distinct parts of the study. If a memory operation fails, the controller keeps the original history rather than applying an incomplete update. Implementation details are in Appendix C.

4.2 PAMER⁺: Step Level Evidence Selection

A summary can cover many steps even when the current request needs only a small part of them. PAMER⁺ records `step_details` for each summary, linking its contents to the original historical steps. A selected summary identifies candidate memory, but does not authorize restoring every step that it covers. The selector specifies `raw_steps_needed`, and only those complete historical blocks enter the recall path. The summary therefore helps locate the relevant memory, while selected raw steps provide the supporting evidence for the current reasoning state. The selected steps must belong to the chosen memory, and invalid selections are rejected. Detailed validation, cache reuse, and fallback rules are given in Appendix C.3.

5 Experiments

We evaluate task performance and token consumption on the full WorkBuddyBench benchmark, compare context-management methods on Pilot40, and examine results across several model backbones. We then inspect context growth and the separate roles of compression and recall.

5.1 Experimental Setup

WorkBuddyBench contains 260 tasks: 80 Code, 50 Office, 60 Security, and 70 Web. Pilot40 contains 10 fixed tasks from each domain. Methods within a comparison use the same task set. The memory-decision corpus is collected separately from these evaluation tasks.

We report mean task reward multiplied by 100. Overall scores average over tasks, rather than giving equal weight to domains of different sizes. Tokens denote mean total input and output tokens per task, including the main agent and auxiliary memory calls. Cache tokens are not added separately because they are already included in input usage. Detailed evaluation protocols, token accounting, and system results are provided in Appendix D.

Table 3: **Full benchmark system performance and token usage.** Scores are multiplied by 100. Green and red indicate favorable and unfavorable changes from the base system.

Model	Code	Office	Sec.	Web	Avg.	Tokens ↓
DeepSeek-V4-Flash	77.0	81.8	47.8	72.1	69.9	2.69M
+ PAMER	75.3 (-1.7)	79.8 (-2.0)	52.3 (+4.5)	70.0 (-2.1)	69.4 (-0.4)	0.72M (-73.2%)
+ PAMER ⁺	76.7 (-0.3)	80.9 (-0.9)	51.6 (+3.8)	76.3 (+4.2)	71.6 (+1.7)	0.76M (-71.7%)

5.2 Main Results

Full benchmark. Table 3 shows the performance and token tradeoff on all 260 tasks. PAMER reduces token usage while retaining a similar overall score. PAMER⁺ improves the reported overall score at a slightly higher token budget than PAMER. The effects differ across domains, so the aggregate result should not be read as an improvement on every type of task.

5.3 Baselines and Cross Model Evaluation

Context-management baselines. Table 4 compares simple controls and published methods on Pilot40. Periodic summarization has a higher average score in the displayed comparison, but uses more tokens. CoMem uses fewer tokens, but has a substantially lower score. PAMER and PAMER⁺ combine relatively strong task performance with a smaller token budget than the base agent.

Table 4: **Comparison with context management methods on WorkBuddyBench Pilot40.** All methods use the same fixed 40 tasks. Scores are multiplied by 100. Parenthesized values show changes from the DeepSeek-V4-Flash base agent.

Method	Code	Office	Sec.	Web	Avg.	Tokens ↓
Base and Ours						
DeepSeek-V4-Flash ⁴⁸	72.9	81.6	44.6	71.0	67.5	1.89M
+ PAMER	76.3 (+3.4)	79.9 (-1.7)	43.9 (-0.7)	72.0 (+1.0)	68.0 (+0.5)	0.90M (-52.4%)
+ PAMER ⁺	71.3 (-1.6)	86.6 (+5.0)	47.5 (+2.9)	70.0 (-1.0)	68.9 (+1.3)	0.98M (-48.1%)
Simple Context Baselines						
Sliding Window ($K = 5$)	66.8 (-6.1)	67.4 (-14.2)	32.3 (-12.3)	59.0 (-12.0)	56.4 (-11.2)	2.95M (+56.3%)
Sliding Window ($K = 10$)	74.4 (+1.4)	73.1 (-8.6)	40.1 (-4.5)	61.0 (-10.0)	62.1 (-5.4)	1.98M (+5.1%)
Sliding Window ($K = 20$)	79.3 (+6.4)	85.1 (+3.5)	40.3 (-4.3)	70.0 (-1.0)	68.7 (+1.1)	2.02M (+7.0%)
Periodic Summary ($n = 3$)	65.4 (-7.5)	67.7 (-14.0)	26.3 (-18.3)	82.7 (+11.7)	60.5 (-7.0)	2.21M (+17.2%)
Periodic Summary ($n = 5$)	83.6 (+10.7)	75.7 (-5.9)	40.0 (-4.5)	81.3 (+10.3)	70.2 (+2.6)	2.37M (+25.5%)
API-based Methods						
PACE ²⁴	70.4 (-2.6)	55.8 (-25.9)	43.5 (-1.0)	64.0 (-7.0)	58.4 (-9.1)	1.10M (-41.6%)
LLMLingua-2 ²¹	64.6 (-8.3)	73.3 (-8.3)	40.1 (-4.5)	70.0 (-1.0)	62.0 (-5.5)	2.40M (+27.3%)
SelfCompact ⁴	76.8 (+3.9)	79.6 (-2.0)	36.2 (-8.4)	71.0 (0.0)	65.9 (-1.6)	1.57M (-17.0%)
ACON-Core ¹⁵	71.8 (-1.1)	68.6 (-13.1)	47.9 (+3.4)	65.0 (-6.0)	63.3 (-4.2)	1.40M (-25.7%)
Self-GC ²⁵	63.5 (-9.5)	71.7 (-10.0)	43.7 (-0.9)	71.0 (0.0)	62.5 (-5.1)	1.66M (-12.4%)
LRE ³³	62.6 (-10.3)	63.4 (-18.2)	27.1 (-17.4)	68.0 (-3.0)	55.3 (-12.2)	2.29M (+21.3%)
CoMem ¹⁰	56.4 (-16.5)	59.9 (-21.8)	2.5 (-42.1)	58.0 (-13.0)	44.2 (-23.3)	0.77M (-59.5%)
SAM ⁹	67.6 (-5.3)	81.6 (0.0)	47.2 (+2.7)	67.0 (-4.0)	65.9 (-1.7)	1.35M (-28.7%)
SWE-Pruner ²³	63.5 (-9.5)	72.6 (-9.0)	40.9 (-3.7)	65.0 (-6.0)	60.5 (-7.1)	1.50M (-20.5%)
Sculptor ⁵	51.2 (-21.7)	81.6 (0.0)	35.2 (-9.3)	69.0 (-2.0)	59.3 (-8.3)	1.29M (-31.5%)
Released-policy Method						
ACM ¹¹	38.7 (-34.2)	55.9 (-25.7)	6.2 (-38.4)	34.0 (-37.0)	33.7 (-33.8)	2.36M

Different model backbones. Table 5 shows reduced token consumption, but model-dependent quality changes. PAMER⁺ improves the scores of MiMo-V2.5 and Qwen3.8-Flash, while GPT-5.6-Luna loses performance under both variants. The results support applicability across different task-executing backbones, while the memory controller uses the same frozen Qwen3.5-9B feature extractor across these evaluations. Additional cross-model usage details are reported in Appendix D.3.

Table 5: **Cross-model evaluation on WorkBuddyBench Pilot40.** Scores are multiplied by 100. All models are evaluated on the same fixed 40 tasks, with 10 tasks per domain. Parenthesized values show changes from the corresponding base agent. Bold indicates the best overall score and lowest token consumption within each backbone.

Model	Code	Office	Sec.	Web	Avg.	Tokens ↓
GPT-5.6-Luna	44.8	72.1	55.7	56.0	57.2	676K
+ PAMER	54.4 (+9.6)	57.8 (-14.3)	52.2 (-3.5)	55.0 (-1.0)	54.9 (-2.3)	81K (-88.0%)
+ PAMER ⁺	41.4 (-3.4)	57.7 (-14.4)	55.3 (-0.4)	66.0 (+10.0)	55.1 (-2.1)	97K (-85.7%)
GLM-5.3-Flash ⁴⁹	43.2	85.0	66.1	61.7	64.0	762K
+ PAMER	44.4 (+1.2)	86.5 (+1.5)	70.7 (+4.6)	66.7 (+5.0)	67.1 (+3.1)	318K (-58.3%)
+ PAMER ⁺	39.6 (-3.6)	84.4 (-0.6)	70.7 (+4.6)	60.0 (-1.7)	63.7 (-0.3)	310K (-59.3%)
HY-3	49.8	80.6	59.5	46.0	59.0	1441K
+ PAMER	53.6 (+3.8)	64.8 (-15.8)	64.2 (+4.7)	51.0 (+5.0)	58.4 (-0.6)	420K (-70.9%)
+ PAMER ⁺	43.6 (-6.2)	80.0 (-0.6)	67.7 (+8.2)	55.0 (+9.0)	61.6 (+2.6)	348K (-75.9%)
MiMo-V2.5 ⁵⁰	32.4	78.4	45.9	55.0	52.9	1114K
+ PAMER	44.8 (+12.4)	65.9 (-12.5)	75.0 (+29.1)	61.0 (+6.0)	61.7 (+8.8)	276K (-75.2%)
+ PAMER ⁺	59.0 (+26.6)	71.0 (-7.4)	75.0 (+29.1)	70.0 (+15.0)	68.8 (+15.8)	282K (-74.7%)
Qwen3.8-Flash ⁵¹	76.5	85.0	62.9	85.0	77.4	1207K
+ PAMER	77.6 (+1.1)	74.8 (-10.2)	79.8 (+16.9)	80.0 (-5.0)	78.1 (+0.7)	869K (-28.0%)
+ PAMER ⁺	80.1 (+3.6)	83.9 (-1.1)	72.6 (+9.7)	100.0 (+15.0)	84.2 (+6.8)	874K (-27.6%)

5.4 Ablation and Online Behavior

Average token consumption does not reveal how context changes within a long interaction. Figure 3 combines the memory operations, component results, and context trajectories. Compression repeatedly reduces the active history, after which context grows as new interaction occurs. Its benefit can therefore persist across later requests that would otherwise resend the removed history.

In the reported component results, removing compression raises token usage, whereas removing recall lowers task reward. These patterns are consistent with complementary roles: compression controls accumulated history, and recall restores information that has left the active context. Appendix E.2 gives a trajectory. The component batch is separate from the Pilot40 baseline comparison. Additional figure interpretation and reproducibility details are provided in Appendix E.

6 Conclusion

Memory needs can be predicted from language model states before the next action. The examined hidden representations contain information beyond context length and interaction metadata, while compression and recall exhibit distinct patterns across model depth. Recent context preserves much of this predictive signal, whereas selected historical evidence recovers information that recency alone cannot retain. These findings motivate PAMER and PAMER⁺, which combine state guided compression with selective access to external memory. Across the reported evaluations, the resulting systems consistently reduce context consumption, while changes in task performance

remain dependent on the model and domain. Overall, our results connect the internal prediction of memory needs with practical context management for long horizon language model agents.

AI Use Statement

Generative AI tools were used for language editing and literature retrieval and discovery, including identifying potentially relevant papers, search keywords, and related work, as well as surveying the current state of research on relevant topics. An LLM was also used as an annotator to identify compression and recall needs in the memory-decision corpus, as described in the paper and appendix. The annotation procedure and resulting labels were reviewed by the authors. The final selection, interpretation, and presentation of the literature and experimental results were determined by the authors. All AI-assisted content was reviewed by the authors, who take full responsibility for the final content of this work.

Reproducibility Statement

We provide detailed descriptions of the data construction, question-grouped train/validation/test splits, probe architectures and evaluation protocols, context construction, retrieval procedures, controller configuration, token accounting, and benchmark task sets in the main paper and appendices. Hyperparameters and implementation-specific settings used by PAMER and PAMER⁺ are reported where applicable, together with the fixed Pilot40 task manifest and the definitions of the reported evaluation metrics. These details are intended to support independent reproduction of the reported analyses and system evaluations.

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A Memory-Decision Data and Probe Analysis

This appendix details the memory-decision corpus, representation probes, compact-context analysis, online controller, and supplementary evaluations. The mathematical results describe the stated losses and control rules. Probing results measure predictive information, rather than identify a causal circuit used by the language model.

A.1 Annotated Memory Decisions and Decision-State Reconstruction

The analysis corpus contains 72,912 decision points from 2,521 trajectories covering 600 questions. These questions are separate from the WorkBuddyBench evaluation tasks.

An LLM annotator identifies decision states where compression or recall is needed. For trajectory τ , let $\mathcal{A}^c(\tau)$ and $\mathcal{A}^r(\tau)$ denote the annotated compression and recall decision sets. At decision step t , with o_t denoting the context available before the next action, the labels are

$$y_t^m = \mathbf{1}[t \in \mathcal{A}^m(\tau)], \quad m \in \{c, r\}. \quad (11)$$

Here c denotes compression and r denotes recall. Compression indicates that older context should be condensed while preserving task-relevant information. Recall indicates that the current reasoning state needs earlier evidence that is no longer available in the active context. Recall is evaluated only when valid external memory exists. A trajectory may contain multiple positive decision points.

Representation extraction uses the provenance-matched OPD iter-2 Qwen3.5-9B checkpoint, which remains frozen during probe fitting. The annotator output and the future action are used only to define the target and are not included in the representation input.

All states and trajectory replicas with the same `query_id` remain in one split. Training, validation, and testing use grouped 70/15/15 partitions with seeds 0–4. The seed controls both the grouped split and readout initialization. Within-trajectory states remain statistically dependent.

A.2 Probe Architecture and Training

The primary feature is the final input-token state after the output RMSNorm of frozen Qwen3.5-9B, $\mathbf{h}_t \in \mathbb{R}^{4096}$. In online use, this feature extractor is separate from the task-executing model; the controller does not require hidden-state access to an API-backed acting model. Each target has an independent readout:

$$\begin{aligned} p_{t,\text{lin}}^m &= \sigma((\mathbf{w}^m)^\top \mathbf{h}_t + b^m), \\ p_{t,\text{mlp}}^m &= \sigma((\mathbf{v}^m)^\top \text{GELU}(\mathbf{W}^m \mathbf{h}_t + \mathbf{b}^m) + c^m). \end{aligned} \quad (12)$$

The MLP hidden width is 256. Including biases, the linear and MLP probes have 4,097 and 1,049,089 parameters per target. Only readout parameters are trained; the language model remains frozen.

The Full Context reference used in the compact-context analysis is the **faithful-hint MLP** with hidden width 256. It retains the recorded context-token hint in the serialized input. Table 6 reports its test means. The same configuration supplies the Full Context teacher and reference in the compact-context analysis. The online compression heads are the separately trained Recent2 decision-distillation students, not the Full Context teacher.

This MLP has the highest reported F1 for both targets among the hint/capacity settings in Table 8; it does not maximize every ranking metric. In particular, the faithful-hint linear compression probe has

Table 6: **Full Context reference for the compact-context analysis.** Five-seed mean test metrics for the faithful-hint MLP; F1 uses per-seed validation-selected thresholds. Values are rounded only for display.

Target	AUROC $\uparrow$	AUPRC $\uparrow$	F1 $\uparrow$
Compression	0.829	0.690	0.639
Recall	0.780	0.597	0.558

higher AUROC. Probe training updates only the readout. The recorded MLP defaults are AdamW, 50 epochs, learning rate 10^{-3} , batch size 256, and hidden width 256. The checkpoint is chosen by validation label loss; the classification threshold is then chosen by validation F1.

For each training split,

$$\mathcal{L}_m = -\frac{1}{N_m} \sum_{i=1}^{N_m} [\rho_m y_i^m \log p_i^m + (1 - y_i^m) \log(1 - p_i^m)], \quad \rho_m = \frac{N_m^-}{N_m^+}. \quad (13)$$

The threshold for a fitted probe is selected by validation F1 and held fixed on the test set:

$$\hat{\theta}_{m,s} \in \arg \max_{\theta \in \Theta} \text{F1}_{\text{val},s}(\mathbf{1}\{p_t^m \geq \theta\}, y_t^m). \quad (14)$$

AUROC is the primary ranking metric. AUPRC complements it for imbalanced targets. Classification recall is distinct from the memory operation called recall. For positive and negative scores, empirical AUROC is

$$\widehat{\text{AUROC}} = \frac{1}{N^+ N^-} \sum_{i:y_i=1} \sum_{j:y_j=0} \left[\mathbf{1}\{z_i > z_j\} + \frac{1}{2} \mathbf{1}\{z_i = z_j\} \right]. \quad (15)$$

A.3 Properties of the Weighted Readout

For $p = \sigma(z)$, the per-example loss has the stable form

$$\ell_\rho(z, y) = (\rho y + 1 - y) \log(1 + e^z) - \rho y z, \quad \frac{\partial \ell_\rho}{\partial z} = (\rho y + 1 - y)p - \rho y. \quad (16)$$

The logarithm is evaluated as $\max(z, 0) + \log(1 + e^{-|z|})$ for numerical stability.

Proposition A.1 (Population score under class weighting). *Let $\eta(\mathbf{h}) = \Pr(Y = 1 \mid \mathbf{h})$ and $\rho > 0$. For $0 < \eta < 1$, conditional weighted BCE has the unique minimizer*

$$p_\rho^*(\mathbf{h}) = \frac{\rho \eta}{1 - \eta + \rho \eta}, \quad \text{logit } p_\rho^* = \text{logit } \eta + \log \rho. \quad (17)$$

Proof. The conditional risk is $R(p) = -\rho \eta \log p - (1 - \eta) \log(1 - p)$. Solving $R'(p) = -\rho \eta / p + (1 - \eta) / (1 - p) = 0$ gives the stated minimizer. Since $R''(p) = \rho \eta / p^2 + (1 - \eta) / (1 - p)^2 > 0$, it is unique. Taking log odds gives the second identity. The inverse relation is

$$\eta = \frac{p_\rho^*}{\rho - (\rho - 1)p_\rho^*}. \quad (18)$$

Boundary cases follow by limits. □

The weighting changes the probability interpretation of the optimal score. It does not certify calibration of a finite fitted probe. Validation-based threshold selection therefore remains part of the protocol.

A.4 Observable Controls, Probe Capacity, and Memory Hints

The controls measure whether length, interaction progress, tool metadata, or surface text explain the memory-decision signal. Recent2 text controls use a shorter input than the Full Context hidden-state probes. Their comparison is not a matched-token-budget experiment.

Table 7: **Memory-decision prediction controls.** Diagnostic rows show mean $\pm$ reported standard deviation over five question-grouped splits. The faithful-hint MLP reference is listed separately with mean metrics.

Signal	Compression		Recall	
	AUROC $\uparrow$	AUPRC $\uparrow$	AUROC $\uparrow$	AUPRC $\uparrow$
Length + turn	0.751 $\pm$ 0.066	0.533 $\pm$ 0.053	0.552 $\pm$ 0.074	0.336 $\pm$ 0.036
Last tool type	0.562 $\pm$ 0.026	0.335 $\pm$ 0.054	0.527 $\pm$ 0.056	0.315 $\pm$ 0.043
Last action type	0.583 $\pm$ 0.019	0.346 $\pm$ 0.057	0.535 $\pm$ 0.056	0.319 $\pm$ 0.039
Length + turn + tool type	0.758 $\pm$ 0.056	0.543 $\pm$ 0.054	0.549 $\pm$ 0.058	0.361 $\pm$ 0.066
Recent2 TF-IDF	0.616 $\pm$ 0.027	0.416 $\pm$ 0.072	0.610 $\pm$ 0.056	0.398 $\pm$ 0.075
Recent2 text embedding	0.546 $\pm$ 0.015	0.363 $\pm$ 0.047	0.594 $\pm$ 0.041	0.385 $\pm$ 0.086
Full Context (faithful-hint MLP)	0.829	0.690	0.780	0.597

The Full Context faithful-hint MLP reference outperforms the tested observable controls for both targets. Table 8 reports additional probe-capacity and memory-hint measurements. The no-hint setting removes explicit memory-token hints, whereas the faithful setting preserves the recorded hint.

A.5 Layerwise Diagnostic Measurements

The layerwise study uses independently fitted no-hint linear readouts on Qwen3.5-9B to compare depths and pooling locations. The faithful-hint MLP used as the Full Context reference in the compact-context analysis is reported separately and is not part of the layerwise comparison. The final linear diagnostic is retained so that a same-capacity pooling comparison remains available.

At layer ℓ , independently trained probes use either the last input state or mean pooling:

$$\mathbf{h}_{t,\text{last}}^{(\ell)} = \phi^{(\ell)}(o_t)[-1], \quad \mathbf{h}_{t,\text{mean}}^{(\ell)} = \frac{1}{n_t} \sum_{j=1}^{n_t} \phi^{(\ell)}(o_t)[j]. \quad (19)$$

At every intermediate checkpoint with both pooling measurements, the last-token linear readout outperforms mean pooling. Recall reaches its highest listed intermediate-layer mean at block 16, while compression AUROC increases across the listed intermediate last-token checkpoints.

Table 8: **Additional probe-capacity and memory-hint measurements.** Mean AUROC, AUPRC, and validation-threshold F1 over five splits. Each row denotes the named probe and hint configuration.

Setting	AUROC $\uparrow$	AUPRC $\uparrow$	F1 $\uparrow$
Compression			
No hint, linear	0.831	0.670	0.601
No hint, MLP	0.827	0.665	0.606
Faithful hint, linear	0.836	0.697	0.624
Faithful hint, MLP	0.829	0.690	0.639
Recall			
No hint, linear	0.765	0.585	0.526
No hint, MLP	0.763	0.592	0.541
Faithful hint, linear	0.774	0.598	0.543
Faithful hint, MLP	0.780	0.597	0.558

Table 9: **Layerwise linear diagnostics and the compact-context reference.** Diagnostic rows show mean $\pm$ reported standard deviation over five question-grouped splits. The bottom row uses the faithful-hint MLP reference from the compact-context analysis and is not part of a same-capacity layer comparison.

Representation	Pooling	Compression		Recall	
		AUROC $\uparrow$	AUPRC $\uparrow$	AUROC $\uparrow$	AUPRC $\uparrow$
Block 4	Last	0.757 ± 0.051	0.563 ± 0.063	0.722 ± 0.058	0.526 ± 0.087
	Mean	0.706 ± 0.027	0.483 ± 0.063	0.632 ± 0.073	0.435 ± 0.079
Block 8	Last	0.799 ± 0.044	0.623 ± 0.082	0.768 ± 0.053	0.594 ± 0.112
	Mean	0.735 ± 0.044	0.493 ± 0.068	0.690 ± 0.058	0.483 ± 0.093
Block 16	Last	0.810 ± 0.031	0.644 ± 0.082	0.780 ± 0.054	0.583 ± 0.108
	Mean	0.762 ± 0.048	0.547 ± 0.064	0.741 ± 0.048	0.529 ± 0.099
Block 24	Last	0.814 ± 0.024	0.649 ± 0.089	0.765 ± 0.061	0.577 ± 0.116
	Mean	0.756 ± 0.049	0.534 ± 0.047	0.726 ± 0.059	0.515 ± 0.100
Block 32	Last	0.824 ± 0.032	0.659 ± 0.087	0.743 ± 0.088	0.560 ± 0.118
	Mean	0.708 ± 0.059	0.478 ± 0.069	0.692 ± 0.045	0.474 ± 0.066
Final RMSNorm	Last	0.831 ± 0.036	0.670 ± 0.100	0.765 ± 0.067	0.584 ± 0.106
	Mean	0.712 ± 0.058	0.479 ± 0.074	0.696 ± 0.047	0.479 ± 0.067
<i>Compact-context Full Context reference: faithful-hint MLP</i>					
Final RMSNorm	Last	0.829	0.690	0.780	0.597

The compact-context Full Context reference is reported separately and is not used to claim a matched readout-capacity comparison with every diagnostic row. The final state is used for online compression; the recall probe remains an analysis target, not a retrieval gate.

Proposition A.2 (Logit separation is not a ranking metric). *For $z = \mathbf{w}^\top \mathbf{h} + b$ with finite class-conditional means,*

$$\Delta = \mathbb{E}[z \mid Y = 1] - \mathbb{E}[z \mid Y = 0] = \mathbf{w}^\top (\boldsymbol{\mu}_+ - \boldsymbol{\mu}_-). \quad (20)$$

Replacing z by $z' = az + c$, $a > 0$, preserves AUROC and changes the gap to $\Delta' = a\Delta$.

Proof. Linearity of expectation gives the displayed gap; the intercept cancels. For each positive–negative pair, $z'_i > z'_j$ if and only if $z_i > z_j$, and ties are also preserved. Every summand in Equation (15) is consequently unchanged. Taking the difference of conditional expectations of $az + c$ gives $\Delta' = a\Delta$. $\square$

Layerwise logit gaps are therefore interpreted together with ranking performance, not as scale-free measures of semantic information. The capacity-control and layerwise tables preserve their respective source aggregations: for example, no-hint linear recall AUPRC is 0.584743 in the capacity-control export and 0.584336 in the layerwise export. Their three-decimal displays, 0.585 and 0.584, are not two roundings of one identical underlying aggregate.

A.6 Paired Compact-Context Comparisons

For paired split-level differences $d_s = M_{A,s} - M_{B,s}$ and $\bar{d} = S^{-1} \sum_s d_s$, the reported two-sided sign-flip calculation is

$$p_{\text{flip}} = 2^{-S} \sum_{\epsilon \in \{-1,1\}^S} \mathbf{1} \left\{ \left| S^{-1} \sum_s \epsilon_s d_s \right| \geq |\bar{d}| \right\}. \quad (21)$$

The inferential interpretation requires sign exchangeability. With $S = 5$, the observed sign pattern and its global negation have the same absolute statistic, so $p_{\text{flip}} \geq 2/2^5 = 0.0625$. The five-split analysis does not establish a two-sided $p < 0.05$ result under this calculation.

Table 10: **Paired compact-context differences across split seeds.** C and R denote compression and recall; positive differences favor the first method. The listed tests concern compact-context interventions and do not reuse significance results from an earlier main-reference configuration.

Comparison	Target / metric	Mean difference	Sign-flip p
Recent2 KD – Recent2	C / AUROC	+0.0058	0.500
	R / AUROC	−0.0080	0.500
	C / F1	+0.0570	0.0625
	R / F1	+0.0278	0.3125
Raw Top5 – Hidden Bank Recent5	C / AUROC	+0.0627	0.0625
	R / AUROC	+0.0389	0.1875

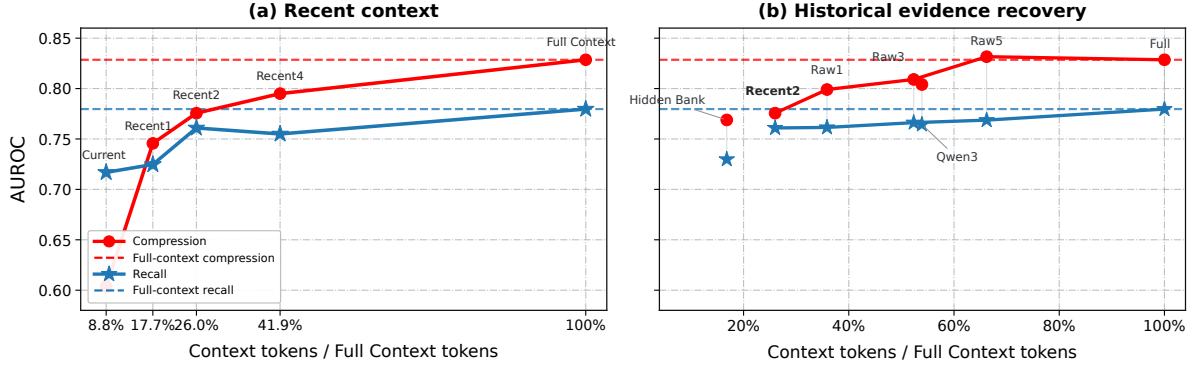


Figure 4: **Decision information at different input budgets.** Left: recent context. Right: historical evidence recovery. Token fractions refer to the corresponding offline comparison.

B Compact Context and Historical Evidence

B.1 Context Construction and Evaluation

Let P_t be the system and task prefix, $\mathcal{R}_t$ the two most recent complete interaction blocks, and $\mathcal{E}_t$ retrieved historical evidence. The principal constructions are

$$C_t^{\text{recent}} = P_t \oplus \mathcal{R}_t, \quad C_t^{\text{recover}} = P_t \oplus \mathcal{E}_t \oplus \mathcal{R}_t. \quad (22)$$

The prefix and annotation labels are held fixed when changing history. The Current condition is identified as *prefix only* in the saved results. It therefore supplies a no-row-history reference; the paper does not treat it as proof of a causal contribution of the task prefix. Other conditions add the specified recent blocks or historical evidence. Relative input cost on the evaluated states is

$$B(C) = 100 \frac{\sum_i \text{Tok}(C_i)}{\sum_i \text{Tok}(C_i^{\text{full}})}. \quad (23)$$

This ratio measures offline text input, not full-task token consumption or the storage cost of a hidden-state bank. Agreement with the Full Context teacher is a separate measurement:

$$\text{Agr}_m = N_m^{-1} \sum_i \mathbf{1}\{\hat{y}_{i,\text{candidate}}^m = \hat{y}_{i,\text{full}}^m\}. \quad (24)$$

Agreement does not replace evaluation against the annotation labels.

Recent2 uses 26.0% of the full textual input and retains much of the memory-decision signal. Additional history is useful in some, but not all, forms: selected raw evidence outperforms the evaluated hidden-bank alternatives, while recurrent and scalar-state additions do not consistently improve over Recent2. Token budgets differ across these conditions and are reported alongside predictive performance.

B.2 Decision Distillation

For student logit z , annotation label y , and fixed Full Context teacher score $q \in [0, 1]$, the compression heads use

$$\ell_{\text{KD}}(z, y, q) = \ell_{\rho}(z, y) + \lambda \text{BCE}(\sigma(z), q), \quad \lambda = 0.5. \quad (25)$$

Table 11: **Compact-context and historical-evidence comparison.** C and R denote compression and recall; dots denote unreported teacher agreement.

Representation	Tokens(%)	Compression		Recall		Agreement	
		AUROC	F1	AUROC	F1	Comp.	Recall
Full Context	100.0	0.829	0.639	0.780	0.558	.	.
Recent context							
Current	8.8	0.604	0.451	0.717	0.532	.	.
Recent1	17.7	0.746	0.520	0.725	0.524	.	.
Recent2	26.0	0.776	0.541	0.761	0.520	.	.
Recent4	41.9	0.795	0.550	0.755	0.555	.	.
Scalar state and decision distillation							
Recent2 + scalar state	26.0	0.767	0.569	0.746	0.552	.	.
Recent4 + scalar state	41.9	0.792	0.579	0.757	0.586	.	.
Recent2 + decision KD	26.0	0.781	0.598	0.753	0.548	0.819	0.840
Recent4 + decision KD	41.9	0.794	0.571	0.758	0.530	0.824	0.857
Recovered raw evidence							
Selective Raw Top1	35.9	0.799	0.569	0.761	0.542	0.826	0.817
Selective Raw Top3	52.3	0.809	0.565	0.766	0.554	0.826	0.834
Selective Raw Top5	66.2	0.832	0.602	0.769	0.567	0.859	0.865
Hidden and recurrent memory							
Hidden Bank Uniform5	16.8	0.722	0.526	0.729	0.540	0.731	0.779
Hidden Bank Recent5	16.8	0.769	0.529	0.730	0.520	0.785	0.744
Hidden Bank Similarity1	16.8	0.716	0.514	0.713	0.507	0.729	0.721
Hidden Bank Similarity3	16.8	0.722	0.509	0.703	0.510	0.729	0.722
Hidden Bank Similarity5	16.8	0.728	0.555	0.712	0.493	0.722	0.747
Hidden Bank Similarity8	16.8	0.725	0.552	0.703	0.535	0.758	0.736
GRU512	16.8	0.745	0.499	0.727	0.513	0.773	0.757
Hybrid GRU + Similarity5	16.8	0.734	0.517	0.672	0.508	0.734	0.684

The teacher is used only during training. The logit gradient is

$$\partial_z \ell_{\text{KD}} = (\rho y + 1 - y + \lambda) \sigma(z) - \rho y - \lambda q. \quad (26)$$

Proposition B.1 (Distillation as weighted soft-label learning). *For $\rho > 0$, $\lambda \geq 0$, and binary y , define*

$$a = \rho y + 1 - y + \lambda, \quad \tilde{y} = \frac{\rho y + \lambda q}{a}. \quad (27)$$

Then $\ell_{\text{KD}} = a \text{BCE}(\sigma(z), \tilde{y})$. Moreover, conditional on the student’s compact representation h , let $\eta(h) = \Pr(Y = 1 \mid h)$ and $\mu(h) = \mathbb{E}[q \mid h]$. The optimal unrestricted student probability is

$$p_\lambda^*(h) = \frac{\rho \eta(h) + \lambda \mu(h)}{1 + (\rho - 1) \eta(h) + \lambda}. \quad (28)$$

Proof. Writing $p = \sigma(z)$, the coefficient of $-\log p$ in the combined loss is $\rho y + \lambda q$, and that of $-\log(1 - p)$ is $1 - y + \lambda(1 - q)$. Their sum is $a > 0$, and their ratio to a gives the soft target. After conditioning on h , the coefficients become $A = \rho\eta + \lambda\mu$ and $B = 1 - \eta + \lambda(1 - \mu)$. Minimizing $-A \log p - B \log(1 - p)$ gives $p^* = A/(A + B)$, yielding Equation (28). When $A, B > 0$, strictly positive second derivative proves uniqueness. Endpoint solutions follow by continuity. The conditional excess risk is

$$R_h(p) - R_h(p_\lambda^*) = (A + B) \text{KL}(\text{Bern}(p_\lambda^*) \| \text{Bern}(p)), \quad (29)$$

which follows directly by subtracting the two cross-entropies. $\square$

Distillation transfers the teacher’s conditional score signal into the compact representation. It does not require the teacher to be perfectly calibrated, nor imply that every metric improves. Empirically, compression F1 improves more than compression AUROC, while recall AUROC decreases in the Recent2 KD condition.

B.3 Decision and Ranking Stability

Proposition B.2 (Stability under compact-context perturbations). *Apply the same fixed linear readout $z = \mathbf{w}^\top \mathbf{h} + b$ to full and compact representations. Suppose $\|\mathbf{h}_f - \mathbf{h}_c\|_2 \leq \varepsilon$ and put $\delta = \|\mathbf{w}\|_2 \varepsilon$. For a probability threshold $0 < \theta < 1$ and $a = \text{logit } \theta$, the two decisions agree whenever*

$$|z_f - a| > \delta. \quad (30)$$

If the score bound holds almost surely under the evaluated state distribution (including the empirical distribution on fixed examples), then

$$\Pr(\hat{Y}_f \neq \hat{Y}_c) \leq \Pr(|Z_f - a| \leq \delta). \quad (31)$$

For fixed sets containing $N^+ > 0$ positive and $N^- > 0$ negative examples, it also implies

$$|\widehat{\text{AUROC}}_f - \widehat{\text{AUROC}}_c| \leq \frac{1}{N^+ N^-} \sum_{i:y_i=1} \sum_{j:y_j=0} \mathbf{1}\{|z_{f,i} - z_{f,j}| \leq 2\delta\}. \quad (32)$$

Proof. Cauchy–Schwarz gives $|z_f - z_c| \leq \|\mathbf{w}\|_2 \|\mathbf{h}_f - \mathbf{h}_c\|_2 \leq \delta$. A logit farther than δ from a cannot cross the threshold, proving the decision statements. For any positive–negative pair, the difference between its compact and full score gaps is at most 2δ . A full gap with absolute value greater than 2δ therefore retains its sign. Its AUROC summand is unchanged. Every remaining pair can change that summand by at most one, including ties. Summing these bounds and dividing by $N^+ N^-$ proves Equation (32). $\square$

These sufficient conditions concern the same readout and a stated representation bound. They are not numerical guarantees for the separately trained probes in the experimental tables.

B.4 Retrieval Controls and Evidence Budgets

The offline retrieval study compares saved raw historical blocks. Selective Raw Top K uses token-frequency cosine similarity between the current last block and eligible older raw blocks, excluding the retained Recent2 blocks. This lexical selector is not a learned oracle and does not use future

actions or labels to rank the evidence. Qwen conditions instead use Qwen3-Embedding-8B and the recorded task/Recent2 query serialization. Offline Qwen retrieval returns raw text, whereas the canonical online PAMER controller retrieves stored summaries. The two settings have different retrieval units; offline raw-block results do not establish an online summary-retrieval effect.

Table 12: **Offline retrieval controls.** Token percentages are relative to Full Context; agreement is measured against the corresponding teacher.

Retriever	Tokens(%)	AUROC		Agreement	
		Comp.	Recall	Comp.	Recall
Random Top3	47.2	0.793	0.752	0.786	0.849
Recency Top3	46.2	0.792	0.753	0.818	0.857
Selective Raw Top1	35.9	0.799	0.761	0.826	0.817
Selective Raw Top3	52.3	0.809	0.766	0.826	0.834
Selective Raw Top5	66.2	0.832	0.769	0.859	0.865
Qwen Top1	37.8	0.794	0.753	0.793	0.839
Qwen Top3	53.9	0.804	0.764	0.838	0.838
Qwen Top5	67.7	0.816	0.775	0.834	0.872

Increasing retrieval cardinality changes both evidence and input cost. Top3 is an intermediate operating point, not a measured maximizer of AUROC. For complete-block costs $\ell_j \geq 0$, an evidence budget obeys

$$\sum_{j \in \mathcal{J}_t} \ell_j \leq B_{\text{evidence}}. \quad (33)$$

The prefix, recent context, and serialization overhead are additional parts of the final request.

Table 13: **Qwen Top3 evidence budgets.** Mean, P95, and maximum describe recalled evidence tokens, not total prompt length.

Variant	Recalled tokens			Agreement	
	Mean	P95	Max	Comp.	Recall
Top3 cap 8K	5,540	8,021	8,192	0.818	0.835
Top3 cap 16K	8,765	14,801	16,383	0.843	0.837
Top3 uncapped	9,167	16,509	32,245	0.838	0.838

The 16K cap retains agreement close to the uncapped condition, whereas the 8K cap reduces agreement. These are offline evidence-budget measurements, not a sweep of online task reward.

C Online Memory Controller

C.1 Compression Voting and Deferred Retrieval

After removing duplicate blocks already represented in stored memories, write the working history as $C_t^{\text{eff}} = P_t \oplus \mathcal{O}_t \oplus \mathcal{R}_t$. Five Recent2 decision-distillation MLP heads read the final normalized Qwen3.5-9B state of the serialized $P_t \oplus \mathcal{R}_t$ input. The saved probe system prompt and context-token hint are applied before serialization. With per-seed validation thresholds θ_s ,

$$v_{t,s} = \mathbf{1}\{p_{t,s}^c \geq \theta_s\}, \quad u_t = \mathbf{1}\{\text{Tok}(\mathcal{O}_t) \geq 1024\} \mathbf{1}\left\{\sum_{s=1}^5 v_{t,s} \geq 3\right\}. \quad (34)$$

On successful compression, an entry stores summary s_j , its embedding, and protocol-aware identities $\mathcal{B}_j$ of the covered raw blocks:

$$M_j = (s_j, \mathbf{e}_j, \mathcal{B}_j), \quad \mathbf{e}_j = E(s_j). \quad (35)$$

The encoder is Qwen3-Embedding-8B with 1,024 output dimensions, distinct from the 4,096-dimensional probe feature. Raw blocks leave active context only after successful storage. The same request receives neither the new summary nor an immediate recall; its reduced context is $P_t \oplus \mathcal{R}_t$.

On a later request, the task and Recent2 define an embedding query. For nonzero embeddings,

$$a_{t,j} = \frac{\mathbf{q}_t^\top \mathbf{e}_j}{\|\mathbf{q}_t\|_2 \|\mathbf{e}_j\|_2}, \quad \mathcal{I}_t = \text{TopK}_j(a_{t,j}), \quad K = 3. \quad (36)$$

At most the available number of memories is retrieved. Query text is serialized from the task and Recent2 and limited to its final 30,000 characters. Selected summaries are inserted after the prefix and before *all* retained raw blocks, not in place of unarchived older blocks:

$$C_t^{\text{recall}} = P_t \oplus \mathcal{E}_t \oplus \mathcal{O}_t \oplus \mathcal{R}_t. \quad (37)$$

Here $\mathcal{E}_t$ contains the selected summaries. The learned recall probe does not gate this branch. A failed probe skips compression but may still use existing memory; a successful compression request never performs recall on that same request.

C.2 Stability of Voting and Retrieval

Proposition C.1 (Majority voting without an independence assumption). *At an eligible state with binary target Y , let E_s indicate an incorrect decision by head s . The five-head majority error satisfies*

$$\Pr(E_{\text{maj}} \mid \mathcal{C}) \leq \min \left\{ 1, \frac{1}{3} \sum_{s=1}^5 \Pr(E_s \mid \mathcal{C}) \right\} \quad (38)$$

for any common conditioning information $\mathcal{C}$. Further, if at least three heads retain their original majority vote under an input perturbation, the majority decision is unchanged.

Proof. A wrong majority requires at least three wrong heads, hence pointwise $\mathbf{1}\{E_{\text{maj}}\} \leq \frac{1}{3} \sum_s \mathbf{1}\{E_s\}$. Taking conditional expectations gives the probability bound. For stability, three unchanged votes already constitute a majority, regardless of the remaining two votes. In particular, a head with logit z_s retains its vote if $|z_s - \text{logit } \theta_s|$ exceeds its maximum logit perturbation. $\square$

Algorithm 1 Online context transformation with PAMER

Require: Canonical request C_t^{raw} , persisted memory $\mathcal{M}_t$, frozen Qwen3.5-9B, five Recent2 KD heads and thresholds.

```

1: if the session identifier or conversation/tool structure is invalid then
2:   | return  $C_t^{\text{raw}}$ 
3: end
4: Load valid memory; if its saved state is corrupt, discard the invalid state and start with empty
   memory
5: Construct  $C_t^{\text{eff}}$  by removing raw blocks already covered by valid persisted memories
6: Split  $C_t^{\text{eff}}$  into  $P_t$ ,  $\mathcal{O}_t$ , and  $\mathcal{R}_t$ 
7: Set  $u_t \leftarrow 0$ 
8: if  $\text{Tok}(\mathcal{O}_t) \geq 1024$  then
9:   | Extract the final Qwen3.5-9B state and evaluate the five heads
10:  | if feature extraction and head evaluation succeed then
11:    | Set  $u_t \leftarrow 1\{\sum_s 1[p_{t,s} \geq \theta_s] \geq 3\}$ 
12:  | else
13:    | Record the probe error and leave  $u_t = 0$  // Existing recall remains possible
14:  | end
15: end
16: if  $u_t = 1$  then
17:   | Generate a summary and its embedding, and validate protocol-aware source identities
18:   | Require successful storage and a strict reduction in active request tokens
19:   | if all transaction checks succeed then
20:     | Commit the memory and return  $P_t \oplus \mathcal{R}_t$  // No same-request recall
21:   | else
22:     | Commit no partial memory and return  $C_t^{\text{raw}}$ 
23:   | end
24: end
25: if  $\mathcal{M}_t \neq \emptyset$  then
26:   | Retrieve at most three summaries using the task and Recent2
27:   | Construct  $C'_t = P_t \oplus \mathcal{E}_t \oplus \mathcal{O}_t \oplus \mathcal{R}_t$ 
28:   | if retrieval and final tool-protocol validation succeed then
29:     | return  $C'_t$ 
30:   | else
31:     | return  $C_t^{\text{raw}}$ 
32:   | end
33: end
34: return  $C_t^{\text{eff}}$ 

```

The bound allows dependent head errors and is not a claim that the ensemble must outperform every individual head.

Proposition C.2 (Top- K retrieval stability). *Let scores a_j and perturbed scores a'_j satisfy $|a'_j - a_j| \leq \delta$ for every memory entry. If there are more than K entries and the ordered-score boundary satisfies*

$$a_{(K)} - a_{(K+1)} > 2\delta, \quad (39)$$

then the retrieved Top- K set is unchanged. For unit-normalized memory embeddings and fixed entries, a query change from $\hat{\mathbf{q}}$ to $\hat{\mathbf{q}}'$ permits $\delta = \|\hat{\mathbf{q}}' - \hat{\mathbf{q}}\|_2$.

Proof. For an originally selected entry i and an unselected entry j , $a_i - a_j \geq a_{(K)} - a_{(K+1)}$. Therefore, $a'_i - a'_j \geq a_i - a_j - 2\delta > 0$. No unselected entry can overtake any selected entry, proving set equality. For cosine scores with unit memory vector $\hat{\mathbf{e}}_j$, Cauchy–Schwarz gives $|a'_j - a_j| = |\langle \hat{\mathbf{q}}' - \hat{\mathbf{q}}, \hat{\mathbf{e}}_j \rangle| \leq \delta$. $\square$

This result characterizes score-margin robustness, not the semantic sufficiency of the retrieved memories.

C.3 Step-Level Evidence in PAMER⁺

PAMER⁺ denotes the step-granular evidence-selection extension. The supplied implementation snapshot places this component in the LLM-router integration. The canonical embedding sidecar described above retrieves whole summaries and does not itself import the exact-step component. The component algorithm below must therefore not be read as evidence that every canonical PAMER run executed that extension.

The router is shown the current Recent2 context and the structured summary index, including `step_details`; it selects summary identifiers and the smallest requested `raw_steps_needed` set. The target future action is not an input. Summary identifiers locate candidate memories but do not authorize expansion of every covered step. Only complete raw blocks explicitly named by the selector are supplied to the Recall Agent. The pure step-granular component validates and normalizes this selection; it does not itself implement a compression gate or invoke a model. For requested identifiers $\mathcal{I}_t$,

$$\mathcal{I}_t \subseteq \bigcup_{j \in \mathcal{I}_t} \mathcal{B}_j. \quad (40)$$

Partitions are checked for gaps, overlaps, missing steps, and invalid identifiers. Selecting a summary does not authorize restoration of every step it covers. Cache identity uses sorted summary and step identifiers; reuse assumes unchanged associated content and configuration. The key is computed from sorted summary and raw-step identifiers, rather than the router’s goal wording or unrelated history updates. The recorded cache protocol distinguishes an empty selection, an invalid packet, selector changes, material changes, and a hit. A successful packet or a declined recall may be reused for an unchanged exact selector.

The caller distinguishes a valid empty selection from a failed operation. An empty selection injects no recalled text. Invalid or failed recall results are not injected and do not authorize deletion of raw history. The calling controller applies its operation-specific fallback. Final conversation and tool-structure validation is required before a transformed request is sent to the agent. The source snapshot includes additional router, retry, and request-scaffold differences; its historical system comparisons do not isolate the effect of exact-step selection.

C.4 Storage, Evidence Selection, and Token Accounting

Proposition C.3 (Successful storage precedes removal). *Suppose a compression update removes active raw blocks only after the summary, embedding, and valid provenance links are stored*

Algorithm 2 Exact-step selection validation in the PAMER⁺ component**Require:** Stored memories, selected summary identifiers $\mathcal{I}_t$, requested raw step identifiers $\mathcal{J}_t$.**Ensure:** Validated recalled evidence, a valid empty selection, or failure $\perp$.

```

1: if  $\mathcal{J}_t = \emptyset$  then
2:   | return an empty evidence set
3: end
4: Validate the partitions and provenance links of the selected memories
5: if the links are invalid then
6:   | return  $\perp$ 
7: end
8: Select the complete raw blocks specified by  $\mathcal{J}_t$ 
9: if a step is missing, duplicated, or unauthorized then
10:  | return  $\perp$ 
11: end
12: Compute cache identity from the sorted summary and step identifiers
13: if a cached result exists for this selection then
14:  | return the cached result
15: end
16: Run the recall operation using only the selected evidence
17: if the recall call fails or the result cites steps outside  $\mathcal{J}_t$  then
18:  | return  $\perp$ 
19: end
20: Cache the validated result
21: return the recalled evidence

```

successfully, and a failed update returns the original request. Then no failed update removes information from the returned active request. If raw-step validation enforces Equation (40), any raw block supplied to the recall operation comes from an authorized memory.

Proof. Before the storage condition holds, the algorithm has no branch returning the reduced context. Failure instead returns the preserved original request, proving the first statement. In the recall branch, valid source links are checked before raw blocks are selected. Invalid selections are rejected. Every surviving selected block consequently has an identifier in the authorized union, proving the second statement. $\square$

Structural validity and provenance do not guarantee that a summary retains all facts or that selected evidence is sufficient for answering the task. Under additive complete-block accounting, selecting $\mathcal{J}_t \subseteq \mathcal{U}_t := \bigcup_{j \in \mathcal{I}_t} \mathcal{B}_j$ reduces the raw evidence volume by

$$\sum_{j \in \mathcal{U}_t} \ell_j - \sum_{j \in \mathcal{J}_t} \ell_j = \sum_{j \in \mathcal{U}_t \setminus \mathcal{J}_t} \ell_j \geq 0. \quad (41)$$

This concerns raw evidence volume; selection calls and serialization can add their own costs.

For a fixed request sequence, let L_j be an archived block's length and m_j the number of avoided occurrences in future requests. Let A_{add} include additional counted memory-operation tokens and

retrieval text. Then

$$\Delta T_{\text{history}} = \sum_j m_j L_j - A_{\text{add}}. \quad (42)$$

Each avoided occurrence contributes L_j , giving the first sum; subtracting added material proves the identity. Complete online runs can have different actions and lengths, so task totals are measured directly. Recent2 and Top3 specify numbers of blocks, not a hard total context limit.

C.5 Implementation Settings and Failure Handling

Summary-source limits are in characters, while model input and output limits are in tokens. If a source exceeds approximately 40,000 characters, messages receive coverage from their beginning and end. Retrieved entries use a stable identifier order, and similarity scores are not inserted into the prompt.

Table 14: **Controller settings.** Character budgets, token budgets, and embedding dimensions are distinct quantities.

Parameter	Value
Compression controller	
Retained complete blocks	2
Feature extractor	Frozen, provenance-matched Qwen3.5-9B
Feature dimension	4,096
Compression heads	5 Recent2 decision-distillation MLPs
Vote rule	3 of 5
Decision thresholds	Validation best F1 threshold, selected independently for each seed
Probe representation	Final token representation after RMSNorm
MLP hidden width (compact-context reference and online heads)	256
Input and output budgets	
Probe maximum input	35,000 tokens
Prefix retained under truncation	2,048 tokens
Output reserve used in the context hint	16,384 tokens
Minimum compression candidate	1,024 tokens
Summary and retrieval	
Summary source budget	Approximately 40,000 characters
Summary model	Qwen3.5-9B (default), temperature 0, thinking disabled
Summary maximum output	1,024 tokens
Summary timeout / attempts	100 seconds / 8
Embedding timeout / attempts	30 seconds / 8
Embedding model	Qwen3-Embedding-8B
Embedding dimension	1,024
Recall cardinality	Top3

Table 15: **Memory-operation fallback rules.**

Condition	Required behavior
Missing session identifier	Return the original request and record the condition.
Unsafe tool protocol	Return the original request without applying a memory transformation.
Candidate below 1,024 tokens	This is not a failure: skip compression; recall from existing memory remains possible.
Probe failure or OOM	Skip compression and record the error; existing-memory recall may still be attempted.
Summary, embedding, or storage failure	Return the original request, commit no incomplete memory, and remove no raw block.
Recall transport or shape failure	Return the original request rather than a partially rewritten context.
Invalid step selection or unsafe injection	Reject the recall result; the canonical controller returns the original request on unsafe injection. The step component authorizes no raw-history deletion.
Valid empty step selection	Inject no recalled evidence; this is not a failed memory operation.
Corrupt state file	Discard invalid saved memory and continue with an empty memory store; do not fabricate prior memories.

D Evaluation Protocol and System Results

D.1 Benchmark and Task Aggregation

All full-benchmark comparisons use the same fixed 260 WorkBuddyBench tasks: 80 Code, 50 Office, 60 Security, and 70 Web. All Pilot40 comparisons use the same fixed 40 tasks, with ten tasks from each domain. No method-specific task filtering is applied.

For task rewards r_i , the aggregate used in this paper is

$$\bar{r} = N^{-1} \sum_{i=1}^N r_i = \sum_d \frac{N_d}{N} \bar{r}_d, \quad \text{Score} = 100\bar{r}. \quad (43)$$

Let $S_d = 100\bar{r}_d$ denote a domain score. For complete measurements on the stated fixed sets, the corresponding aggregates are

$$\begin{aligned} S_{260} &= \frac{80S_{\text{Code}} + 50S_{\text{Office}} + 60S_{\text{Security}} + 70S_{\text{Web}}}{260}, \\ S_{40} &= \frac{S_{\text{Code}} + S_{\text{Office}} + S_{\text{Security}} + S_{\text{Web}}}{4}. \end{aligned} \quad (44)$$

Because Pilot40 contains exactly ten tasks from each of the four domains, the overall score is the equal-weight average of the four domain scores. All averages and score changes are computed before final display rounding. Domain scores and overall scores are rounded separately for presentation, so recomputing an overall score from the displayed one-decimal domain values may differ slightly from the reported task-level aggregate. The presentation retains graded rewards and token usage, without redundant full-pass fractions in score cells.

The recorded outer agent limit is `CBC_MAX_TURNS=256`; this number is a turn limit, not a task count. Tool calls, transport retries, and internal context operations are counted separately. ACM uses a released policy rather than the same API-backed acting model.

D.2 Token and Context Measurements

With counted main-agent and auxiliary calls $\mathcal{C}_i$ for task i ,

$$T_i = \sum_{c \in \mathcal{C}_i} (I_{ic} + O_{ic}), \quad \bar{T} = N^{-1} \sum_i T_i, \quad \delta_T = 100 \left(\frac{\bar{T}_{\text{method}}}{\bar{T}_{\text{base}}} - 1 \right). \quad (45)$$

Cached input is already included in input tokens; reasoning output already included in output usage is not counted again. Auxiliary operations are counted once. Local feature extraction and embedding computation remain additional computational work, not zero-cost operations merely because API token counts do not cover them.

Pilot40 token usage is computed from the recorded input and output totals over the fixed 40 tasks, with per-task usage obtained by dividing the aggregate total by 40. Percent changes are computed from the unrounded totals before display rounding.

Full260 token usage is aggregated over the fixed 260 tasks. The DeepSeek-V4-Flash baseline contains 698,167,232 total tokens, corresponding to approximately 2.69M tokens per task.

For per-turn active context L_{it} , the mean task peak is

$$\bar{P} = N^{-1} \sum_i \max_t L_{it}. \quad (46)$$

It differs from cumulative task tokens. Monetary cost further depends on cached-input, uncached-input, and output prices, so token reduction is not identified with an equal percentage reduction in billed cost.

D.3 Cross-Model Scores on Fixed Pilot40

All cross-model comparisons use the same fixed Pilot40 task set. GPT-5.6-Luna, GLM-5.3-Flash, HY-3, MiMo-V2.5, and Qwen3.8-Flash are each evaluated with the base agent, PAMER, and PAMER⁺ on all 40 tasks, with ten tasks per domain. No model- or method-specific task filtering is applied. Overall scores are computed from the underlying task-level rewards before display rounding.

The controller uses the same frozen Qwen3.5-9B feature extractor when the task-executing model changes. Cross-model deployment therefore does not require transferring a probe between different acting-model hidden spaces.

D.4 Component Ablation

The component ablation is evaluated in a separate execution batch and is reported at the aggregate level.

Removing recall lowers reward from 0.7000 to 0.6370, while removing compression increases tokens per task from 0.899M to 3.180M. These patterns are consistent with the complementary roles of context reduction and historical access, without establishing a universal causal effect.

Table 16: **Cross-model results on the fixed Pilot40 task set.** Scores are multiplied by 100. All variants use the same 40 tasks, with ten tasks per domain. Avg. is the equal-weight average of the four domain scores; Δ is the change from the corresponding base agent.

Backbone	Variant	Code	Office	Sec.	Web	Avg.	Δ
GPT-5.6-Luna	Base agent	44.8	72.1	55.7	56.0	57.2	0.0
	PAMER	54.4	57.8	52.2	55.0	54.9	-2.3
	PAMER ⁺	41.4	57.7	55.3	66.0	55.1	-2.1
GLM-5.3-Flash	Base agent	43.2	85.0	66.1	61.7	64.0	0.0
	PAMER	44.4	86.5	70.7	66.7	67.1	+3.1
	PAMER ⁺	39.6	84.4	70.7	60.0	63.7	-0.3
HY-3	Base agent	49.8	80.6	59.5	46.0	59.0	0.0
	PAMER	53.6	64.8	64.2	51.0	58.4	-0.6
	PAMER ⁺	43.6	80.0	67.7	55.0	61.6	+2.6
MiMo-V2.5	Base agent	32.4	78.4	45.9	55.0	52.9	0.0
	PAMER	44.8	65.9	75.0	61.0	61.7	+8.8
	PAMER ⁺	59.0	71.0	75.0	70.0	68.8	+15.8
Qwen3.8-Flash	Base agent	76.5	85.0	62.9	85.0	77.4	0.0
	PAMER	77.6	74.8	79.8	80.0	78.1	+0.7
	PAMER ⁺	80.1	83.9	72.6	100.0	84.2	+6.8

Table 17: **Cross-model token usage on fixed Pilot40.** K denotes thousands of tokens. Totals include the counted main-agent and auxiliary memory calls. Changes are relative to the corresponding base agent.

Backbone	Variant	Tokens/task (K) ↓	Change (%)
GPT-5.6-Luna	Base agent	676	0.0
	PAMER	81	-88.0
	PAMER ⁺	97	-85.7
GLM-5.3-Flash	Base agent	762	0.0
	PAMER	318	-58.3
	PAMER ⁺	310	-59.3
HY-3	Base agent	1441	0.0
	PAMER	420	-70.9
	PAMER ⁺	348	-75.9
MiMo-V2.5	Base agent	1114	0.0
	PAMER	276	-75.2
	PAMER ⁺	282	-74.7
Qwen3.8-Flash	Base agent	1207	0.0
	PAMER	869	-28.0
	PAMER ⁺	874	-27.6

Table 18: **Component ablation in a separate execution batch.** Reward uses the original scale.

Variant	Reward $\uparrow$	Tokens/task $\downarrow$	Reported peak $\downarrow$
Base agent	0.6868	3.066M	280.1K
PAMER	0.7000	0.899M	97.5K
w/o Recall	0.6370	0.924M	107.1K
w/o Compression	0.6887	3.180M	253.3K

Table 19: **Selected events in the illustrative trajectory.**

Turn	Recorded operation	Evidence in the saved trace
19	manage_context	41 messages are summarized into memory 1.
28	manage_context	19 messages are summarized into memory 2.
36	query_memory	The agent requests memory 1 during continued search.
47	manage_context	39 messages are summarized into memory 3.
71	manage_context	49 messages are summarized into memory 4.
87	get_document	Document 20026 is retrieved as candidate supporting evidence.
93	Final answer	The agent returns Tom Rice, matching the recorded gold answer.

E Trajectory Analysis and Evaluation Tasks

E.1 Context Dynamics

The full-benchmark context-growth comparison uses the same fixed 260-task list for all displayed methods. Let $L_{i,m,t}$ be active input tokens for task i , method m , and turn t . The per-turn mean is

$$\bar{L}_{m,t} = \frac{1}{|\mathcal{I}_{m,t}|} \sum_{i \in \mathcal{I}_{m,t}} L_{i,m,t}, \quad \mathcal{I}_{m,t} = \{i \in \mathcal{Q}_{260} : L_{i,m,t} \text{ is observed at turn } t\}. \quad (47)$$

A task that has already ended contributes no later-turn context value, so the active trajectory count can decrease with turn even though the underlying evaluation set is the same fixed 260 tasks for every method. The 128K line is an analysis reference, not a controller-enforced limit.

E.2 Compression and Recall in One Trajectory

Question 558 is a multi-document identification task combining a museum founder, a magazine editor, and biographical clues. The final answer, Tom Rice, matches the task’s recorded answer. The trajectory contains 94 turns, 81 searches, 16 document retrievals, four compression calls, and two memory queries. Selected events show how old evidence becomes accessible again after leaving active context. This single trace is a qualitative illustration of memory operations, not an additional WorkBuddyBench evaluation cohort or a change to the fixed task sets.

E.3 Fixed Pilot40 Task Identifiers

The following identifiers define the fixed Pilot40 task set, with ten tasks per domain. The same 40-task manifest is used for the Pilot40 method comparison and the cross-model evaluation. The full-benchmark evaluation uses the fixed 260-task WorkBuddyBench list.

Table 20: Fixed Pilot40 task identifiers.

Domain	Index	Task identifier
Code	1	api_contract-hard-markup_errors
	2	api_contract-hard-openapi_params
	3	api_contract-hard-token_errors
	4	api_contract-hard-validation_errors
	5	bug_fix-easy-a_crash_in_local
	6	bug_fix-easy-filtered_relation_queryset_arg
	7	bug_fix-easy-invalid_filterwarnings_regex_error
	8	bug_fix-medium-error_key_uses_data_key
	9	bug_fix-medium-errors_from_earlier_indices
	10	bug_fix-medium-incorrect_lineno_on_fstring
Office	11	analyst-forecast-extract-L3-018
	12	api-usage-explain-cli-l3-001
	13	board-material-update-timeline-excel
	14	calendar-dida-sync-state
	15	channel-period-compare-L4-017
	16	cloudagent-sdk-doc-validation-report
	17	contract-extract-L3-014
	18	cross-week-dashboard-migration
	19	crypto-backtest-chain-L4-002
	20	daily-creation-checkpoint-recovery
Security	21	agent-to-agent-injection-hard-multistep
	22	apt-multi-source-correlation-hard-multistep
	23	bb-bin-dns-parse-010
	24	bb-bin-firmware-audit-007
	25	bb-bin-format-log-004
	26	bb-bin-int-length-005
	27	bb-bin-ipc-cache-001
	28	bb-bin-media-parse-008
	29	bb-bin-oob-read-003
	30	bb-bin-parse-crash-006
Web	31	animated-explainer-L3-028
	32	atmosphere-game-L4-035
	33	blog-editor-draft-recovery-L4-059
	34	browser-clipper-extension-L4-005
	35	canvas-webgl-scene-L4-026
	36	chart-generation-L2-025
	37	checkout-incident-analysis-L4-049
	38	city-article-theme-variants-L4-066
	39	claims-drawer-state-review-report-L3-071
	40	cohort-retention-dashboard-L4-054